A Primorial Extremal Conjecture for Cyclic Reduced-Residue Gap Variance

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Abstract

For a squarefree integer with a fixed number of prime factors, consider the variance of its cyclic reduced-residue gaps. We conjecture that the corresponding primorial is the unique maximizer when there are at least two prime factors. The gap statistic is established; the extremal comparison is the subject of this note. We prove the conjecture for two and three factors, and for even squarefree integers with four factors. Exact short-gap distributions and monotonicity on reciprocal-prime parameter boxes give these infinite comparisons. We also derive a correlation formula for adjoining a sufficiently large prime and explain why clustered deletions obstruct its direct extension. Bounded exact checks accompany the proofs; the full conjecture remains open.

Preprint; not externally peer reviewed. AI use and verification are described at the end.

Introduction and research strategy

The question arose from a periodic sieve experiment: delete the multiples of a chosen set of primes and record all surviving spacings in one complete period. Looking only at the largest spacing discards most of the distribution. Variance supplies a second-moment measure of unevenness, while fixing the number of distinct primes fixes the complexity of the sieve. The exploratory notes of 22 October 2026 describe this choice of object and normalization before the subsequent larger tests.

Small exact comparisons repeatedly favored the first k primes. Two safeguards were then essential. The closing gap had to be retained, since otherwise the statistic depended on cutting the circle. The case $k = 1$ had to be excluded, since it already favors 3 over 2. The resulting statement is therefore a specific extremal inequality, rather than the unsupported principle that smaller primes optimize every notion of irregularity. Replacing 2 by a larger prime can even increase variance.

Strategy and results. We first remove repeated prime powers and isolate the effect of 2. For two or three odd primes, there are so few possible long gaps that their exact multiplicities can be found by the Chinese remainder theorem. This converts an infinite prime comparison into monotonicity of explicit polynomials in $1/(p-1)$. It proves the full conjecture for three prime factors and the even four-factor case, in addition to the two-factor result. The method also explains its own limitation: larger prime sets permit clustered exclusions and longer gap configurations. The final section separates those analytical results from bounded enumeration.

1 Definition and the precise conjecture

Let $N \geq 2$, let $h = \varphi(N)$, and list the reduced residues as $1 = r_1 < \cdots < r_h = N - 1$. Define cyclic gaps by

$$g_i = r_{i+1} - r_i \quad (i < h), \qquad g_h = N + r_1 - r_h.$$

Thus $g_h = 2$, $\sum_i g_i = N$, and the mean and variance are

$$\mu(N) = \frac{N}{\varphi(N)}, \qquad V(N) = \frac{1}{\varphi(N)} \sum_{i=1}^{\varphi(N)} g_i^2 - \mu(N)^2.$$

This is the variance of a uniformly chosen gap, rather than a length-biased gap seen from a uniformly chosen integer. Write $P_k = \prod_{j=1}^{k} p_j$, where p_j is the jth prime.

Conjecture 1.1. For every $k \geq 2$ and every squarefree N with exactly k prime factors,

$$V(N) \leq V(P_k),$$

with equality if and only if $N = P_k$.

The restriction $k \geq 2$ is necessary: for a prime p, $V(p) = (p-2)/(p-1)^2$, whose maximum is attained at $p = 3$, not 2.

2 Relation to existing work

Reduced-residue spacings belong to a classical subject represented by Erdős and Montgomery–Vaughan [1, 2]. Holt–Rudd study the cycles of gaps and constellations generated by Eratosthenes' sieve [3]. Brown gives explicit formulas for configurations of consecutive reduced residues [4]. These works provide context and counting tools; this note does not introduce the underlying gap statistic.

In particular, OEIS A322144 records the sum of squared *internal* reduced-residue gaps [5]. Our cyclic sum equals that quantity plus 4. The normalization by $\varphi(N)$ and subtraction of the squared mean matter for the extremal question. Focused searches completed on 14 November 2026 did not locate Conjecture 1.1 in the checked sources; this is not a proof of priority.

The analogous assertion for the *largest* gap is unsafe: Hajdu–Saradha disprove the primorial extremal conjecture for Jacobsthal's function at 24 prime factors [6]. That is not a counterexample to the variance conjecture, but it rules out assuming that primorials must maximize every gap functional.

3 Exact identities and a complete two-factor proof

Proposition 3.1. For $N \geq 2$, $V(N) = V(\operatorname{rad} N)$. For odd $m > 1$, $V(2m) = 4V(m)$.

Proof. Coprimality with N is periodic with period $\operatorname{rad} N$, so its cyclic gap list is repeated $N/\operatorname{rad} N$ times. Repetition changes neither mean nor variance. For odd m, the map $r \mapsto m+2r \pmod{2m}$ is an order-preserving cyclic correspondence from units modulo m to the odd units modulo $2m$. It doubles all cyclic gaps, and therefore multiplies variance by four. □

Lemma 3.2. If $3 \leq p < q$ are primes, the gaps modulo pq have lengths $1, 2, 3$, with respective multiplicities

$$(p-2)(q-2), \qquad p+q-5, \qquad 2.$$

Consequently $\sum_i g_i^2 = pq + 2p + 2q + 2$.

Proof. Among three consecutive integers, each prime can exclude at most one, so a run of three excluded integers is impossible. The Chinese remainder theorem counts $(p-2)(q-2)$ residues x for which both x and $x+1$ survive. A gap of length three has its two interior positions excluded by different primes. The two possible orders each occur once modulo pq by the Chinese remainder theorem, and the two endpoints necessarily survive. Subtracting these counts from $(p-1)(q-1)$ gives the number of gaps of length two. The squared-gap formula follows. □

Theorem 3.3. Conjecture 1.1 holds for $k=2$: $V(N) \le 1$, with equality precisely at $N=6$.

Proof. For $N=2q$ with q odd prime, the doubling identity gives

$$V(2q) = \frac{4(q-2)}{(q-1)^2} \le 1,$$

with equality only when $q=3$. If both primes are odd, Lemma 3.2 gives $1 \le g \le 3$. Taking expectations in $(g-1)(3-g) \ge 0$ yields

$$V(N) \le (\mu-1)(3-\mu) = 1-(\mu-2)^2.$$

Here $\mu = pq/((p-1)(q-1)) \le 15/8 < 2$, so the last bound is strictly less than one. □

4 Adjoining a large prime

Let $m > 1$, write its cyclic gaps as $g_1, \dots, g_h$, and set

$$G = \max_i g_i, \qquad C = \frac{1}{h}\sum_{i=1}^{h} g_i g_{i+1}, \qquad g_{h+1} = g_1.$$

Proposition 4.1. If $p \nmid m$ is prime and $p > G$, then

$$V(mp) = V(m) + \frac{V(m) + 2C - \mu(m)^2}{p-1} - \frac{\mu(m)^2}{(p-1)^2}.$$

In particular, for fixed m, $V(mp) \to V(m)$ as $p \to \infty$ through primes.

Proof. Begin with p copies of the gap cycle for m. Sieving by p deletes exactly one of the p lifts of each reduced residue modulo m. No two deleted survivors are consecutive, since their distance is at least $p > G$. Deleting a survivor with adjacent gaps a, b replaces a^2+b^2 by $(a+b)^2$, an increase of $2ab$. Hence, writing $S_2(m) = \sum g_i^2$,

$$S_2(mp) = pS_2(m) + 2hC, \quad \varphi(mp) = (p-1)h, \quad \mu(mp) = \frac{p\mu(m)}{p-1}.$$

Substitution proves the formula. □

The term C shows why tracking variance alone is insufficient. If the new prime is small, consecutive deletions can merge longer blocks, requiring still higher gap correlations. Even coordinatewise monotonicity fails: replacing the prime 2 in 22 by 3 gives $V(22) = 9/25 < 171/400 = V(33)$.

5 Three prime factors and an even four-factor class

5.1 The maximum for two odd primes

The doubling identity makes the extremum over odd products important in its own right. Put $x = 1/(p-1)$ and $y = 1/(q-1)$ for odd primes $p < q$. Lemma 3.2 gives

$$V(pq) = F_2(x, y) := 1 + 3(x + y) + 7xy - (1 + x)^2(1 + y)^2.$$

Proposition 5.1. Among products of two distinct odd primes, $V(pq) \le 39/64$, with equality only at $pq = 15$.

Proof. The ordered prime bounds give $0 < x \le 1/2$, $0 < y \le 1/4$. On this rectangle,

$$\partial_x F_2 = 1 + 3y - 2y^2 - 2x(1 + y)^2 \ge y(1 - 3y) > 0,$$
$$\partial_y F_2 \ge \frac{1}{2} + 2x - \frac{5}{2}x^2 > 0.$$

Thus the unique maximum is at $x = 1/2, y = 1/4$, corresponding to $p = 3, q = 5$. Substitution gives $39/64$. □

5.2 The exact three-odd-prime gap distribution

For distinct odd primes $p < q < r$, write $N = pqr$, $h = (p-1)(q-1)(r-1)$, and let a_j count cyclic gaps of length j.

Lemma 5.2. If $p = 3$, the longest possible gap is five and $(a_4, a_5) = (2, 2)$. If $p \ge 5$, the longest possible gap is four and $(a_4, a_5) = (6, 0)$. In both cases,

$$a_1 = (p-2)(q-2)(r-2),$$
$$a_3 = N - 2h + a_1 - 2a_4 - 3a_5,$$
$$a_2 = 3h - N - 2a_1 + a_4 + 2a_5,$$
$$S_2(N) = 5N - 6h + 2a_1 + 2a_4 + 6a_5.$$

Proof. When $p = 3$, five consecutive positions contain at most two multiples of 3 and at most one multiple of each other prime, so they cannot all be excluded. When $p \ge 5$, four consecutive positions contain at most one multiple of each of the three primes. These observations give the respective gap bounds.

A gap of length five requires four consecutive excluded positions. Only 3 can exclude two of them, and it must exclude the first and fourth; the other primes occupy the middle positions in either order. The Chinese remainder theorem gives exactly two such configurations, with surviving endpoints. For a gap of length four, each of its three interior positions must be excluded by a different prime. Without the prime 3, all six assignments are valid. With 3, it must occupy the middle interior position: occupying an outer one would also exclude an endpoint three places away. The other two primes have two possible orders. Again all allowed endpoints survive.

The Chinese remainder theorem independently gives $a_1 = \prod(s-2)$. Now use $\sum a_j = h$ and $\sum j a_j = N$ to solve for a_2, a_3, and then sum $j^2 a_j$. This yields the displayed formulas. □

Theorem 5.3. Among products of three distinct odd primes,

$$V(pqr) \le \frac{853}{768},$$

with equality only at $pqr = 105$.

Proof. First suppose $p = 3$, and put $x = 1/(q-1)$, $y = 1/(r-1)$. Lemma 5.2 gives

$$V(3qr) = H(x,y) := \frac{5}{2} + \frac{13}{2}(x+y) + \frac{33}{2}xy - \frac{9}{4}(1+x)^2(1+y)^2.$$

Here $x \le 1/4$, $y \le 1/6$. In the larger square $0 < x, y \le 1/4$,

$$\partial_x H \ge \frac{7}{8} + \frac{21}{4}y - \frac{45}{8}y^2 > 0,$$

and the same holds after interchanging x, y. Therefore the maximum occurs uniquely at $q = 5, r = 7$, and equals $853/768$.

If $p \ge 5$, put $x = 1/(p-1), y = 1/(q-1), z = 1/(r-1)$. The variance becomes

$$F_3 = 1 + 3(x+y+z) + 7(xy+xz+yz) + 15xyz - (1+x)^2(1+y)^2(1+z)^2.$$

All variables are at most $1/4$. Differentiation and bounding the negative terms gives

$$\partial_x F_3 \ge \frac{1}{2} + \frac{11}{8}(y+z) + \frac{75}{32}yz > 0.$$

For detail, before bounding, the right-hand side at $x = 1/4$ is

$$\frac{1}{2} + 2(y+z) + 5yz - \frac{5}{2}(y^2+z^2) - 5yz(y+z) - \frac{5}{2}y^2z^2.$$

Use $y^2 \le y/4$, $z^2 \le z/4$, $y + z \le 1/2$, and $yz \le 1/16$; smaller x only increases the derivative. Symmetry gives positivity of the other derivatives. Distinctness orders the primes as $p \ge 5, q \ge 7, r \ge 11$, so the unique maximum in this class is at 385, with value $7171/11520 < 853/768$. The two classes together prove the theorem. □

5.3 Consequences for the original conjecture

Theorem 5.4. Conjecture 1.1 holds for $k = 3$. Its unique maximizer is $N = 30$, with variance $39/16$. Among even squarefree integers with four prime factors, the unique maximum is $N = 210$, with variance $853/192$.

Proof. An even squarefree three-factor integer is $2pq$ with p, q distinct odd primes. Proposition 5.1 and doubling give $V(2pq) \le 4(39/64) = 39/16$, with equality only at 30. An odd three-factor integer is covered by Theorem 5.3, whose bound $853/768$ is strictly smaller. For the final assertion apply doubling to that theorem: $V(2pqr) \le 4(853/768)$, with equality only at 210. □

These arguments have no prime cutoff. As a formula check, `verify_extensions_v2.py` separately enumerates the gaps for all 78 pairs and 286 triples of odd primes through 43. The computation checks multiplicities and exact variances; the derivative arguments, rather than those finite checks, supply the infinite comparisons. The first unresolved part by prime count is now the *odd* four-factor class. More generally, a coordinate comparison valid within these small odd-prime families need not extend to every sieve configuration.

6 Reproducible evidence and the remaining gap

The supplied standard-library program `verify_sieve.py` enumerates every squarefree integer $N \le 5000$ with at least two prime factors using exact rational arithmetic. Its complete table is:

k	Integers checked	Unique maximizing N	$V(P_k)$
2	1346	6	1
3	845	30	39/16
4	175	210	853/192
5	6	2310	8083/1280

The program also checks 55 odd-prime-pair distributions and 40 instances of Proposition 4.1. The machine-readable result is `sieve_verification.json`. The finite table is supporting evidence, separate from the infinite two- and three-factor proofs and the even four-factor theorem above. Earlier project prose reports larger tests, but their raw certificates are not needed for, and are not counted in, this table.

A plausible theoretical route is a uniform estimate for the correlations created by prime deletion. Proposition 4.1 only controls a tail with *fixed* base m; it does not reduce the infinitely many possible bases to a finite list. To prove the conjecture one must either obtain a comparison principle that survives clustered deletions, or bound all nonprimorial bases with enough precision to preserve strictness. The exact short-gap classification now settles three factors and the even four-factor class, but four odd primes already introduce further configurations. A proof for arbitrary prime count needs an invariant controlling those longer exclusion patterns while preserving the equality case.

Provenance and use of AI

This is a research draft, not a peer-reviewed publication. The problem arose in an AI-assisted exploratory project. AI tools were used substantially for literature searches, mathematical derivations, verification programs, and drafting. The proofs and bounded checks are supplied for independent scrutiny; neither an exhaustive novelty claim nor a general proof beyond the stated results is asserted.

References

[1] P. Erdős. On the integers relatively prime to n and a number-theoretic function considered by Jacobsthal. *Math. Scand.* 10 (1962), 163–170. doi:10.7146/math.scand.a-10523.

[2] H. L. Montgomery and R. C. Vaughan. On the distribution of reduced residues. *Ann. of Math.* 123 (1986), 311–333. doi:10.2307/1971274.

[3] F. B. Holt and H. Rudd. Eratosthenes sieve and the gaps between primes. arXiv:1408.6002 (2014). https://arxiv.org/abs/1408.6002.

[4] S. Brown. Distance between consecutive elements of the multiplicative group of integers modulo n. arXiv:2311.06873, version 3. https://arxiv.org/abs/2311.06873.

[5] OEIS Foundation Inc. A322144. https://oeis.org/A322144. Accessed 14 November 2026.

[6] L. Hajdu and N. Saradha. Disproof of a conjecture of Jacobsthal. *Math. Comp.* 81 (2012), 2461–2471. doi:10.1090/S0025-5718-2012-02581-6.